6.7.1 Different forms of equation of a straight line

A straight line can be uniquely fixed if

- a point on the straight line and the direction of the straight line are given
- two points on the straight line are given

We find equations of a straight line in vector and Cartesian form. To find the equation of a straight line in vector form, an arbitrary point P with position vector $\vec{r}$ on the straight line is taken and a relation satisfied by $\vec{r}$ is obtained by using the given conditions. This relation is called the vector equation of the straight line. A vector equation of a straight line may or may not involve parameters. If a vector equation involves parameters, then it is called a **vector equation in parametric form**. If no parameter is involved, then the equation is called a **vector equation in non – parametric form**.

6.7.2 A point on the straight line and the direction of the straight line are given

(a) Parametric form of vector equation

Theorem 6.11

The vector equation of a straight line passing through a fixed point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$ is $\vec{r} = \vec{a} + t\vec{b}$, where $t \in \mathbb{R}$.

Proof

If $\vec{a}$ is the position vector of a given point A and $\vec{r}$ is the position vector of an arbitrary point P on the straight line, then $\overrightarrow{AP} = \vec{r} - \vec{a}$.

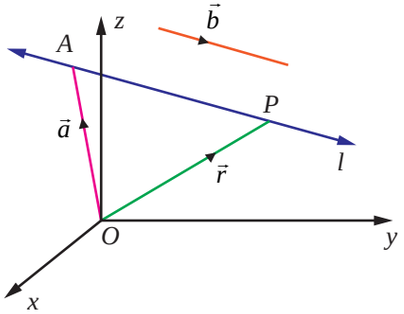


Fig. 6.18

Since $\overrightarrow{AP}$ is parallel to $\vec{b}$, we have

$$\vec{r} - \vec{a} = t\vec{b},\ t \in \mathbb{R} \qquad \ldots (1)$$

$$\text{or} \quad \vec{r} = \vec{a} + t\vec{b},\ t \in \mathbb{R} \qquad \ldots (2)$$

This is the vector equation of the straight line in parametric form.

Remark

The position vector of any point on the line is taken as $\vec{a} + t\vec{b}$.

(b) Non-parametric form of vector equation

Since $\overrightarrow{AP}$ is parallel to $\vec{b}$, we have $\overrightarrow{AP} \times \vec{b} = \vec{0}$

That is, $(\vec{r} - \vec{a}) \times \vec{b} = \vec{0}$.

This is known as the **vector equation** of the straight line in **non-parametric form**.

(c) Cartesian equation

Suppose P is (x, y, z), A is (x_1, y_1, z_1) and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$. Then, substituting $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$ in (1) and comparing the coefficients of $\hat{i}, \hat{j}, \hat{k}$, we get

$$x - x_1 = tb_1, y - y_1 = tb_2, z - z_1 = tb_3 \qquad \ldots (4)$$

Conventionally (4) can be written as

$$\frac{x - x_1}{b_1} = \frac{y - y_1}{b_2} = \frac{z - z_1}{b_3} \qquad \ldots (5)$$

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